



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

TASK 1 REPORT

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Task Experiment 04 of the M-Tech Production & Marketing AI/ML Internship Program, Batch 2026. This task focuses on understanding Machine Learning (ML) fundamentals, its main types, and the end-to-end ML project workflow. M-Tech Production & Marketing, based in Haripur, KPK, provides this training to build real-world concepts, helping us understand how computers learn patterns from data instead of using fixed manual rules.

2. Objectives

- Understand what Machine Learning is and how it differs from traditional programming.
- Learn the three major types of ML: Supervised, Unsupervised, and Reinforcement learning.
- Identify key ML terms like features (inputs) and labels (outputs/targets).
- Learn the step-by-step machine learning workflow and the wider project lifecycle.
- Train a basic regression model using the scikit-learn library.

3. Task Description

The task involves studying how an ML workflow functions from scratch. We need to explain the major steps like data collection, cleaning, splitting, training, and testing. The task requires building a small practical script to train a LinearRegression model that can automatically predict exam scores based on study hours, showing how the computer learns mathematical relationships on its own.

4. Work Done

I completed the lab study and implemented the python practical work successfully. I mapped out the entire 7-step ML workflow and distinguished between features and target labels using concrete examples. I also set up a predictive model using Scikit-Learn in a Jupyter Notebook to calculate future scores for specific study durations and evaluated how changes in input values alter model parameters.

5. Challenges Faced

- **Features vs. Labels:** Finding the exact boundary between features and labels in highly complex datasets can be tricky for beginners.

- **Workflow Depth:** Remembering all the steps of the project lifecycle and understanding why we split data into training and test sets required deep attention.
- **Array Reshaping:** Scikit-learn requires 2D arrays (`.reshape(-1, 1)`) for features even when dealing with a single input variable, which threw minor formatting errors initially.

6. Key Learnings

- **Learning vs. Rules:** In traditional coding, we give rules. In ML, we give data and answers, and the system creates its own rules.
- **Supervised Learning:** Whenever we have labeled data (like area and house price), we use supervised learning. It is divided into regression (predicting continuous numbers) and classification (predicting distinct categories).
- **Workflow Discipline:** Skipping data cleaning or testing on the training data leads to bad performance. Proper evaluation on unseen test data is critical before deploying an ML model.

7. Conclusion

This experiment successfully introduced the basic foundations of Machine Learning workflows. By training a tiny linear regression model, I understood how algorithms optimize internal weights (`coef_`) to match real-world trends. Mastering these foundational definitions and data-splitting principles is essential before moving towards complex neural networks and automated AI systems.

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